

Independent Mathematica Derivation for $D_{s1}/B_{s1} \rightarrow D_s/B_s \gamma$

Ds/Bs gamma project

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Abstract

This note is a clean, independent restart of the symbolic part of the $D_{s1}/B_{s1} \rightarrow D_s/B_s \gamma$ light-cone sum-rule calculation. It records exactly what has been implemented and checked in *Mathematica* and FEYN CALC: the currents, correlator ordering, Wick contraction, propagator routing, Rohrwild three-point/background-field notation, hard-emission traces, Ward identities, E1 projection, inverse-denominator reduction, Feynman parameterization, the first soft two-particle Fierz traces, physical-state mixing, the pure-axial Colangelo limit, and the decay-constant normalization. It does not import the Python implementation and does not yet claim a completed perturbative spectral density or double-Borel sum rule for the tensor current.

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1 Purpose and executable files

The interactive source of this calculation is

```
notebooks/DsBs_gamma_symbolic_derivation.nb.
```

It contains 38 input cells designed to be evaluated from top to bottom or one at a time. The noninteractive regression scripts are

```
scripts/mathematica_current_wick_trace_audit.wl,
scripts/mathematica_correlator_routing_audit.wl,
scripts/mathematica_hard_loop_parameterization.wl,
scripts/mathematica_state_mixing_projection.wl.
```

The first three batch audits write their full machine-readable expressions under `outputs/`; the state-mixing audit prints its exact residuals and interval overlaps directly. This note presents the compact analytic content needed to understand and reproduce those expressions.

2 Conventions and kinematics

We use

$$g_{\mu\nu} = \text{diag}(1, -1, -1, -1), \quad \{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}, \quad (1)$$

$$\sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu], \quad \epsilon^{0123} = +1. \quad (2)$$

The FEYNCalc convention is explicitly checked through `FeynCalc['LeviCivitaSign=-1]`.

The momentum assignment is

$$p' = p + q, \quad (3)$$

where p' is the initial 1^+ momentum, p is the final 0^- momentum, and q is the photon momentum. The photon is real,

$$q^2 = 0, \quad 2p \cdot q = p'^2 - p^2. \quad (4)$$

The heavy flavor is denoted by $Q = c, b$ and its mass by m_Q ; the light flavor is strange with mass m_s .

3 Currents and the Rohrwild correlator

The two basis-current vertices are

$$\Gamma_\mu^A = \gamma_\mu \gamma_5, \quad (5)$$

$$\Gamma_\mu^B = \frac{i}{m_Q + m_s} \sigma_{\mu\alpha} p'^\alpha \gamma_5, \quad (6)$$

and the pseudoscalar vertex is

$$\Gamma_P = i\gamma_5. \quad (7)$$

The tensor vertex is independently represented as

$$\Gamma_\mu^B = -\frac{(\gamma_\mu \not{p}' - \not{p}' \gamma_\mu) \gamma_5}{2(m_Q + m_s)}. \quad (8)$$

Mathematica verifies that the two definitions agree exactly.

The physical-current rotations are kept outside the basis traces:

$$\Gamma_\mu^{\text{low}} = \sin \theta \Gamma_\mu^A + \cos \theta \Gamma_\mu^B, \quad (9)$$

$$\Gamma_\mu^{\text{high}} = \cos \theta \Gamma_\mu^A - \sin \theta \Gamma_\mu^B. \quad (10)$$

Following Rohrwild's vacuum three-point notation [arXiv:0708.1405](#), adapted to a $1^+ \rightarrow 0^- \gamma$ transition, the starting correlator is

$$\Pi_{\mu\nu}^X(p, q) = i^2 \int d^4x d^4y e^{ip \cdot x + iq \cdot y} \langle 0 | T \{ J_P^\dagger(x) j_\nu^{\text{em}}(y) J_\mu^X(0) \} | 0 \rangle, \quad X = A, B, \quad (11)$$

where

$$j_\nu^{\text{em}} = e_s \bar{s} \gamma_\nu s + e_Q \bar{Q} \gamma_\nu Q. \quad (12)$$

Here e_s, e_Q are charges in units of the positive elementary charge e . Introduce the plane-wave background field

$$F_{\alpha\beta}(z) = i(q_\alpha \varepsilon_\beta^* - q_\beta \varepsilon_\alpha^*) e^{iq \cdot z}. \quad (13)$$

The term linear in this field obeys

$$\varepsilon^{*\nu} \Pi_{\mu\nu}^X(p, q) = i \int d^4x e^{ip \cdot x} \langle 0 | T \{ J_P^\dagger(x) J_\mu^X(0) \} | 0 \rangle_F \equiv i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ J_P^\dagger(x) J_\mu^X(0) \} | 0 \rangle, \quad (14)$$

up to the explicitly stated overall QED convention. Thus the former vacuum-to-photon formula was not a different calculation; it is the single-photon component of Eq. (11).

In explicit fields the ordered product is

$$T \{ \bar{s}(x) \Gamma_P Q(x) \bar{Q}(0) \Gamma_\mu^X s(0) \}. \quad (15)$$

Thus the vertex at x carries the final channel p , while the current at the origin carries the initial channel $p' = p + q$.

The current definitions used in the notebook are:

```
gammaA[index_] := GA[index] . GA[5];
gammaB[index_] :=
  I FV[pPrime, alpha] DiracSigma[GA[index], GA[alpha]] . GA[5]/
  (mQ + ms);
gammaBExpanded[index_] :=
  -(GA[index] . GS[pPrime] - GS[pPrime] . GA[index]) . GA[5]/
  (2 (mQ + ms));
gammaP[] := I GA[5];
propNumerator[momentum_, mass_] := GS[momentum] + mass;
```

4 Wick contraction and global signs

For the complete hard contraction, Eq. (15) gives

$$\langle T\{\bar{s}\Gamma_P Q \bar{Q}\Gamma_X s\} \rangle = -\text{Tr}_D[\Gamma_P S_Q \Gamma_X S_s]. \quad (16)$$

The minus sign is the closed-fermion-loop sign. Including the leading correlator factor i , but not yet the propagator or QED-vertex factors, the hard prefactor is therefore

$$-iN_c. \quad (17)$$

If only the heavy flavor is contracted, the strange bilocal remains open and there is no closed fermion loop. The soft contraction begins with $+i$. The Fierz rearrangement

$$\bar{s}_\alpha(x)s_\beta(0) = -\frac{1}{4}\sum_i(\Gamma_i)_{\beta\alpha}\bar{s}(x)\Gamma_i s(0) \quad (18)$$

then gives a soft prefactor $-i/4$ before the free-propagator factor.

In the notebook the bookkeeping is kept visible:

```
fermionLoopSign = -1;
hardWickPrefactor = I fermionLoopSign Nc;
softWickPrefactorBeforeFierz = I;
softWickPrefactorAfterFierz = -I/4;
```

5 Propagator expansion and OPE building blocks

The free momentum-space propagator is

$$S_f^0(r) = \frac{i(\not{r} + m_f)}{r^2 - m_f^2 + i0}. \quad (19)$$

5.1 Light-strange propagator in Rohrwild notation, including m_s

To make the comparison with Rohrwild direct, write $\bar{u} = 1 - u$ and keep the background fields at the point ux . With color indices displayed, the light-strange propagator used as the master expansion is

$$\begin{aligned} s^a(\widehat{x})\bar{s}^b(0) &= \delta^{ab} \left[\frac{i \not{x}}{2\pi^2 x^4} - \frac{m_s}{4\pi^2 x^2} \right] \\ &\quad - \frac{ig_s}{16\pi^2 x^2} \int_0^1 du \{ \bar{u} \not{x} \sigma_{\alpha\beta} + u \sigma_{\alpha\beta} \not{x} \} G_A^{\alpha\beta}(ux) t_{ab}^A \\ &\quad - \frac{ie_s \delta^{ab}}{16\pi^2 x^2} \int_0^1 du \{ \bar{u} \not{x} \sigma_{\alpha\beta} + u \sigma_{\alpha\beta} \not{x} \} F^{\alpha\beta}(ux) \\ &\quad - \frac{m_s}{32\pi^2} \int_0^1 du \left[g_s G_A^{\alpha\beta}(ux) t_{ab}^A + e_s \delta^{ab} F^{\alpha\beta}(ux) \right] \sigma_{\alpha\beta} L_x + \dots, \end{aligned} \quad (20)$$

where

$$L_x = \ln\left(\frac{-x^2\mu^2}{4}\right) + 2\gamma_E. \quad (21)$$

The first line contains only the free propagator and its explicit m_s term. The last line is the linear- m_s Wilson coefficient in a single gluon or electromagnetic background field. No local vacuum-condensate term is written in Eq. (20). Setting m_s and the last line to zero gives precisely the light-propagator form displayed in Eq. (2.12) of Rohrwild. The ellipsis denotes higher-background-field and higher-order mass terms, not local condensate pieces used elsewhere.

Important calculation warning. Although local parts are no longer displayed as part of the propagator, the existing axial numerical sum-rule implementation *does include* the term labeled `heavy_local` in the Python audit. After the double Borel transform it is

$$\Pi_{Q\gamma}^{A,\text{local}} = e_Q e^{-m_Q^2/M^2} \langle \bar{s}s \rangle \left[1 - \frac{m_Q m_s}{M^2} + \frac{m_s^2}{2M^2} \left(1 + \frac{m_Q^2}{M^2} \right) \right]. \quad (22)$$

It has not been removed here, because doing so would change the quoted form factors and widths. The mixed condensate $\langle \bar{s}g_s \sigma \cdot Gs \rangle$ and the vacuum gluon condensate were not included in the radiative three-point calculation.

5.2 Heavy propagator in a background gluon field

For reference, the momentum representation through one background gluon can be organized as

$$S_Q^{ab}(x) = i \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \left\{ \frac{\delta^{ab}(\not{k} + m_Q)}{k^2 - m_Q^2} - \frac{g_s G_{\alpha\beta}^A t_{ab}^A}{4} \frac{\sigma^{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma^{\alpha\beta}}{(k^2 - m_Q^2)^2} + \dots \right\}. \quad (23)$$

For the photon-DA calculation the one-gluon term is used as an operator insertion, rather than replaced by $\langle G^2 \rangle$. The explicit light-cone form used in the symbolic three-particle kernels is

$$S_Q^G(x) = -ig_s \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \int_0^1 du \left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2} G^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha}{m_Q^2 - k^2} G^{\alpha\beta}(ux) \gamma_\beta \right]. \quad (24)$$

Both the $\sigma_{\alpha\beta}$ part and the $x_\alpha \gamma_\beta$ part have Mathematica trace-kernel scripts.

5.3 Electromagnetic background-field propagator

The first-order electromagnetic correction is

$$S_f^\gamma(x) = -ie_f \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \int_0^1 du \left[\frac{\not{k} + m_f}{2(m_f^2 - k^2)^2} F^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha}{m_f^2 - k^2} F^{\alpha\beta}(ux) \gamma_\beta \right]. \quad (25)$$

The overall electromagnetic coupling may either be included in e_f or kept outside the propagator, but not both. In the explicit-vertex trace audit, Eq. (25) is represented equivalently by

$$S_f^0(r)(-iee_f \gamma_\nu \varepsilon^\nu) S_f^0(r+q), \quad (26)$$

with the orientation fixed separately for each diagram. These are alternative representations of the same hard photon-emission term.

5.4 Photon-DA matrix elements

The condensate-dependent photon terms are vacuum-to-photon nonlocal matrix elements. For example,

$$\begin{aligned} \langle \gamma(q, \varepsilon) | \bar{s}(x) \sigma_{\mu\nu} s(0) | 0 \rangle = & -iee_s \langle \bar{s}s \rangle (\varepsilon_\mu q_\nu - \varepsilon_\nu q_\mu) \int_0^1 du e^{i\bar{u}q \cdot x} \left[\chi \phi_\gamma(u) + \frac{x^2}{16} \mathbb{A}(u) \right] \\ & + \text{the twist-4 } h_\gamma \text{ structure,} \end{aligned} \quad (27)$$

and

$$\langle \gamma(q, \varepsilon) | \bar{s}(x) \gamma_\mu s(0) | 0 \rangle = e e_s f_3 \gamma_\mu \tilde{\varepsilon}_\mu \int_0^1 du e^{i\bar{u}q \cdot x} \psi^v(u). \quad (28)$$

The three-particle matrix elements $\langle \gamma | \bar{s}(x) g_s G_{\alpha\beta}(vx) \Gamma s(0) | 0 \rangle$ define $\mathcal{S}, \tilde{\mathcal{S}}, \mathcal{T}_1, \dots, \mathcal{T}_4$. Their axial combination used in the sum rule is

$$\mathcal{F}_1 = \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4 + 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2). \quad (29)$$

6 Momentum routing

6.1 Heavy-line emission

The three fermion momenta are k , $k + q$, and $k - p$. The denominators are

$$d_{H1} = (k + q)^2 - m_Q^2, \quad (30)$$

$$d_{H2} = k^2 - m_Q^2, \quad (31)$$

$$d_{H3} = (k - p)^2 - m_s^2. \quad (32)$$

The current momenta are

$$k - (k - p) = p, \quad (k + q) - (k - p) = p + q. \quad (33)$$

6.2 Strange-line emission

The three momenta are k , $k - p$, and $k - p - q$. The denominators are

$$d_{S1} = k^2 - m_Q^2, \quad (34)$$

$$d_{S2} = (k - p)^2 - m_s^2, \quad (35)$$

$$d_{S3} = (k - p - q)^2 - m_s^2. \quad (36)$$

The corresponding current momenta are

$$k - (k - p) = p, \quad k - (k - p - q) = p + q. \quad (37)$$

This vertex check is important. The old trial route $k - p + q$ gives $k - (k - p + q) = p - q$ at the initial vertex and is therefore not the routing of Eq. (11). It is kept only in the dedicated routing audit as a diagnostic comparison.

7 Hard-emission Dirac traces

Charges, color, denominators, the QED vertex, propagator factors, and the global Wick sign are deliberately kept outside the numerator traces. For a generic basis vertex Γ_μ^X , the heavy-emission numerator is

$$N_{H,\mu\nu}^X = \text{Tr}_D \left[i\gamma_5 (\not{k} + m_Q) \gamma_\nu (\not{k} + \not{q} + m_Q) \Gamma_\mu^X (\not{k} - \not{p} + m_s) \right]. \quad (38)$$

The strange-emission numerator is

$$N_{s,\mu\nu}^X = \text{Tr}_D \left[i\gamma_5 (\not{k} + m_Q) \Gamma_\mu^X (\not{k} - \not{p} - \not{q} + m_s) \gamma_\nu (\not{k} - \not{p} + m_s) \right]. \quad (39)$$

The implementation is short enough to display in full:

```

traceReduce[expression_] :=
  Collect2[
    DiracSimplify[expression, DiracSubstitute67 -> True],
    {mQ, ms}
  ];

heavyTrace[current_] := traceReduce[DiracTrace[
  gammaP[] . propNumerator[k, mQ] . GA[nu] .
  propNumerator[k + q, mQ] . current[mu] .
  propNumerator[k - p, ms]
]];

strangeTrace[current_] := traceReduce[DiracTrace[
  gammaP[] . propNumerator[k, mQ] . current[mu] .
  propNumerator[k - p - q, ms] . GA[nu] .
  propNumerator[k - p, ms]
]];

nAHeavy = heavyTrace[gammaA];
nAStrange = strangeTrace[gammaA];
nBHeavy = heavyTrace[gammaB];
nBStrange = strangeTrace[gammaB];

```

Each of the four traces is recomputed after a cyclic rotation of the matrix chain. All four cyclic differences vanish exactly.

8 Fermion-line Ward identities

For equal masses on the photon-emitting segment and $B = A + q$,

$$(A + m) \not{q} (\not{B} + m) = D_B (A + m) - D_A (\not{B} + m), \quad (40)$$

where $D_A = A^2 - m^2$ and $D_B = B^2 - m^2$. Equation (40) shows explicitly that replacing the photon vertex by q_ν collapses one of the adjacent propagators, as required by the Ward–Takahashi identity.

The notebook checks Eq. (40) independently for the heavy line, with $A = k$ and $B = k + q$, and the strange line, with $A = l - q$ and $B = l$:

```

wardHeavyResidual = FCE[DiracSimplify[
  DiracGammaExpand[
    propNumerator[k, mQ] . GS[q] . propNumerator[k + q, mQ]
    - (SP[k + q, k + q] - mQ^2) propNumerator[k, mQ]
    + (SP[k, k] - mQ^2) propNumerator[k + q, mQ]
  ],
  DiracSubstitute67 -> True, DiracOrder -> True
]] // ScalarProductExpand // Simplify;

```

Both the heavy and strange residuals are zero.

9 E1 projection

The transverse E1 tensor is chosen as

$$S_{\mu\nu} = p_\nu q_\mu - (p \cdot q) g_{\mu\nu}. \quad (41)$$

For $q^2 = 0$,

$$q^\nu S_{\mu\nu} = 0. \quad (42)$$

The normalized projector is

$$\mathcal{P}_{E1}^{\mu\nu} = \frac{p^\nu q^\mu - (p \cdot q) g^{\mu\nu}}{2(p \cdot q)^2}, \quad (43)$$

and satisfies

$$\mathcal{P}_{E1}^{\mu\nu} S_{\mu\nu} = 1. \quad (44)$$

The code is

```
e1Tensor = FV[p, nu] FV[q, mu] - SP[p, q] MT[mu, nu];
e1Projector = e1Tensor/(2 SP[p, q]^2);

e1Project[expression_] := Collect2[
  ScalarProductExpand[
    Contract[expression e1Projector] /. kinematicRules
  ],
  {mQ, ms}
];

projAHeavy = e1Project[nAHeavy];
projAStrange = e1Project[nAStrange];
projBHeavy = e1Project[nBHeavy];
projBStrange = e1Project[nBStrange];
```

The projector normalization, photon Ward contraction, and contraction with a physical initial-state polarization are all checked symbolically.

10 Physical-state projection and the Colangelo limit

10.1 Rotation of the projected invariant amplitudes

Let $\hat{T}_A$ and $\hat{T}_B$ denote the continuum-subtracted, double-Borel E1 invariant amplitudes obtained from the two basis currents. Because the correlator is linear in the current at the initial-state vertex, the physical projections are

$$\begin{pmatrix} \hat{T}_1 \\ \hat{T}_2 \end{pmatrix} = R(\theta) \begin{pmatrix} \hat{T}_A \\ \hat{T}_B \end{pmatrix}, \quad R(\theta) = \begin{pmatrix} \sin \theta & \cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix}, \quad R^T R = \mathbf{1}. \quad (45)$$

The same rotation therefore applies after the E1 projection, after the OPE, and after the Borel transformation; these linear operations commute.

Define the pseudoscalar and physical axial-state residues by

$$\langle 0 | J_P | P(p) \rangle = \frac{f_P m_P^2}{m_Q + m_s}, \quad (46)$$

$$\langle 0 | J_i^\mu | H_i(p', \eta) \rangle = f_i M_i \eta^\mu, \quad i = 1, 2, \quad (47)$$

and use

$$\langle \gamma(q, \varepsilon) P(p) | H_i(p', \eta) \rangle = e G_i [(\varepsilon^* \cdot \eta)(p \cdot q) - (\varepsilon^* \cdot p)(\eta \cdot q)]. \quad (48)$$

The pole term in the projected, double-Borel invariant is then

$$\hat{T}_i^{\text{pole}} = \frac{f_P m_P^2}{m_Q + m_s} f_i M_i G_i \exp\left(-\frac{M_i^2}{\mathcal{M}_1^2} - \frac{m_P^2}{\mathcal{M}_2^2}\right). \quad (49)$$

Consequently the correctly normalized physical coupling is

$$G_i = \frac{m_Q + m_s}{f_P m_P^2 f_i M_i} \exp\left(\frac{M_i^2}{\mathcal{M}_1^2} + \frac{m_P^2}{\mathcal{M}_2^2}\right) \hat{T}_i^{\text{OPE}}. \quad (50)$$

If the basis residues for a given pole are defined with the same mass normalization, Eq. (50) becomes

$$G_1 = \frac{\sin \theta f_A^{(1)} G_A^{(1)} + \cos \theta f_B^{(1)} G_B^{(1)}}{f_1}, \quad (51)$$

$$G_2 = \frac{\cos \theta f_A^{(2)} G_A^{(2)} - \sin \theta f_B^{(2)} G_B^{(2)}}{f_2}. \quad (52)$$

Thus a bare sine–cosine rotation of already normalized couplings is valid only when the corresponding residues cancel.

10.2 Colangelo-style final sum-rule assembly

For the paper we organize the answer at the level of the projected Borel invariants, rather than displaying the unprojected correlation function. In the equal-Borel convention $\mathcal{M}_1^2 = \mathcal{M}_2^2 = 2M^2$, define

$$\mathcal{N}_i(M^2) = \frac{\exp[(M_i^2 + m_P^2)/(2M^2)](m_Q + m_s)}{M_i f_i m_P^2 f_P}, \quad i = 1, 2. \quad (53)$$

The two physical couplings are then presented directly as

$$G_1 = \mathcal{N}_1 \left(\sin \theta \hat{T}_A + \cos \theta \hat{T}_B \right), \quad (54)$$

$$G_2 = \mathcal{N}_2 \left(\cos \theta \hat{T}_A - \sin \theta \hat{T}_B \right). \quad (55)$$

Each basis invariant is decomposed into the same OPE classes,

$$\hat{T}_X = \hat{T}_X^{\text{pert}} + \hat{T}_X^{\langle \bar{s}s \rangle} + \hat{T}_X^{\text{tw}2} + \hat{T}_X^{\text{tw}3} + \hat{T}_X^{\text{tw}4} + \hat{T}_X^{3\text{p}}, \quad X = A, B. \quad (56)$$

For the axial current, the Colangelo-style comparison invariant within the adopted truncation is

$$\begin{aligned} \hat{T}_A = & \int_{(m_Q+m_s)^2}^{s_0} ds e^{-s/M^2} \rho_A^P(s) \\ & + e_Q E_Q \langle \bar{s}s \rangle \left[1 - \frac{m_Q m_s}{M^2} + \frac{m_s^2}{2M^2} \left(1 + \frac{m_Q^2}{M^2} \right) \right] \\ & - e_s \langle \bar{s}s \rangle (E_Q - E_0) M^2 \chi_C \phi_\gamma(u_0) \\ & - e_s \langle \bar{s}s \rangle E_Q \left[-\frac{1}{4} (\mathbb{A}(u_0) - 8\bar{H}_\gamma(u_0)) \left(1 + \frac{m_Q^2}{M^2} \right) - \mathcal{I}_{\mathcal{F}_1}(u_0) \right] \\ & + 2e_s f_{3\gamma} m_Q E_Q \Psi^v(u_0), \end{aligned} \quad (57)$$

where

$$E_Q = e^{-m_Q^2/M^2}, \quad E_0 = e^{-s_0/M^2}, \quad (58)$$

$$\bar{H}_\gamma(u_0) = \int_0^{u_0} du \int_0^u dv h_\gamma(v), \quad (59)$$

$$\Psi^v(u_0) = \int_0^{u_0} du \psi^v(u), \quad (60)$$

and the three-particle contribution is

$$\begin{aligned}\mathcal{I}_{\mathcal{F}_1}(u_0) &= \int_0^{1-u_0} dv \int_0^{u_0/(1-v)} d\alpha_g \mathcal{F}_1(u_0 - (1-v)\alpha_g, 1 - u_0 - v\alpha_g, \alpha_g) \\ &\quad + \int_{1-u_0}^1 dv \int_0^{(1-u_0)/v} d\alpha_g \mathcal{F}_1(u_0 - (1-v)\alpha_g, 1 - u_0 - v\alpha_g, \alpha_g).\end{aligned}\quad (61)$$

Here χ_C is the signed susceptibility convention printed by Colangelo et al. The project stores the positive magnitude $\chi_\gamma = |\chi_C|$, so the twist-2 term is implemented with the equivalent replacement

$$-e_s \langle \bar{s}s \rangle \chi_C \longrightarrow +e_s \langle \bar{s}s \rangle \chi_\gamma. \quad (62)$$

10.3 Paper default: Rohrwild nonlocal transition OPE

The standalone local term in Eq. (57) is *not* part of the paper-default transition sum rule. Denote by $\hat{T}_A^{\text{base}}$ the right-hand side of that equation with the $e_Q \langle \bar{s}s \rangle$ local term removed. The two schemes used in the numerical comparison are

$$\hat{T}_A^{\text{Col}} = \hat{T}_A^{\text{base}} + e_Q E_Q \langle \bar{s}s \rangle \left[1 - \frac{m_Q m_s}{M^2} + \frac{m_s^2}{2M^2} \left(1 + \frac{m_Q^2}{M^2} \right) \right], \quad (63)$$

$$\hat{T}_A^{\text{RW}} = \hat{T}_A^{\text{base}} + \mathcal{I}_A^{\text{em}}, \quad (64)$$

and they are alternatives: the two last terms must never be added.

At the symmetric double-Borel point $u_0 = 1/2$, the current-specific axial trace gives $(1 - 2u)\mathcal{S}_\gamma - \mathcal{T}_4^\gamma$ for the direct part. Both direct integrals vanish by antisymmetry, leaving

$$\mathcal{I}_A^{\text{em}} = e_Q \langle \bar{s}s \rangle (E_Q - E_0) \frac{2(m_P^2 - m_A^2)}{M^2} P_\gamma, \quad P_\gamma = \ln 8 - \frac{1181}{320} = -1.611183458320\dots \quad (65)$$

The source convention used here is

$$\mathcal{S}_\gamma = 60\alpha_g^2(\alpha_q + \alpha_{\bar{q}})[4 - 7(\alpha_q + \alpha_{\bar{q}})], \quad (66)$$

$$\mathcal{T}_4^\gamma = 60\alpha_g^2(\alpha_{\bar{q}} - \alpha_q)[4 - 7(\alpha_q + \alpha_{\bar{q}})]. \quad (67)$$

The third integral in the printed appendix representation of $\mathcal{P}[\mathcal{T}_4^\gamma]$ has lower limit zero and is logarithmically divergent if read literally. Enforcing the double-Borel support $\alpha_{\bar{q}} \geq 1/2$ gives the finite constant P_γ above. This support qualification is retained explicitly in the Python output rather than silently absorbed into a numerical coefficient.

For the tensor basis current, with the stated physical-residue numerator prescription

$$p^2 = m_P^2, \quad p \cdot q = \frac{m_A^2 - m_P^2}{2}, \quad u_0 = \frac{1}{2}, \quad (68)$$

the corresponding nonlocal electromagnetic term is

$$\mathcal{I}_B^{\text{em}} = e_Q \langle \bar{s}s \rangle (E_Q - E_0) \left[(r_0 - r_S) \frac{25}{16} - r_T \frac{5}{16} + 2r_0 \frac{m_P^2 - m_A^2}{M^2} P_\gamma \right], \quad (69)$$

$$r_0 = \frac{m_Q}{m_Q + m_s}, \quad r_S = \frac{m_Q^2 + \frac{1}{2}p \cdot q}{m_Q(m_Q + m_s)}, \quad r_T = \frac{-m_Q^2 + m_P^2 + \frac{3}{2}p \cdot q}{m_Q(m_Q + m_s)}. \quad (70)$$

Equations (65) and (69) are implemented in `scripts/rohrwild_em_da.py`.

Contribution	Rohrwild paper default	Colangelo comparison
Ordinary local condensate in transition LCSR	absent	retained for J_A
$\mathcal{S}_\gamma, \mathcal{T}_4^\gamma$	included nonlocally	not added separately
Gluonic photon DAs	included	included through $\mathcal{F}_1$
Local condensates in two-point f_i sum rules	required	required
Local plus electromagnetic three-particle term	forbidden double counting	forbidden double counting

Table 1: Separation of the local-condensate prescriptions.

For clarity, the perturbative density is written without the ambiguous “ $s \leftrightarrow Q$ ” shorthand. Define

$$\rho_i(s; m_i, m_j, e_i) = -\frac{3e_i}{8\pi^2} \left\{ 2m_i \ln \frac{s - m_j^2 + m_i^2 - \lambda^{1/2}(s, m_j^2, m_i^2)}{s - m_j^2 + m_i^2 + \lambda^{1/2}(s, m_j^2, m_i^2)} + (m_j - m_i) \frac{m_j^2 - m_i^2 - s}{s^2} \lambda^{1/2}(s, m_j^2, m_i^2) \right\}, \quad (71)$$

so that

$$\rho_A^P(s) = \rho_s(s; m_s, m_Q, e_s) - \rho_Q(s; m_Q, m_s, e_Q). \quad (72)$$

The tensor-current invariant is kept in the identical paper-ready decomposition

$$\hat{T}_B = \hat{T}_B^{\text{pert}} + \hat{T}_B^{\langle \bar{s}s \rangle} + \hat{T}_B^{\text{tw}2} + \hat{T}_B^{\text{tw}3} + \hat{T}_B^{\text{tw}4} + \hat{T}_B^{\text{3p}}. \quad (73)$$

The Python result now fills these slots using the diagonal double-dispersion mapping for the perturbative term and the documented physical-residue prescription for the tensor numerators. The ordinary local-condensate slot is zero by the Rohrwild scheme, while Eq. (69) supplies the nonlocal electromagnetic contribution. The next Mathematica stage is an independent cell-by-cell duplication of these already fixed Python prescriptions; it does not change Eqs. (54) and (55).

10.4 Exact pure-axial reduction to Colangelo et al.

Colangelo, De Fazio, and Ozpineci use the single axial current $J_\mu^A = \bar{Q}\gamma_\mu\gamma_5 s$ in their correlator. This is the Hermitian-conjugate orientation of our A current and has the same invariant amplitude. In the convention of Eq. (45), their $D_{s1}(2460)$ calculation is obtained from the low channel by choosing

$$\boxed{\theta = \frac{\pi}{2} = 90^\circ}, \quad J_1 = J_A, \quad J_2 = -J_B. \quad (74)$$

At this point

$$\hat{T}_1 = \hat{T}_A, \quad f_1 = f_A, \quad G_1 = G_A. \quad (75)$$

For the equal-Borel convention $\mathcal{M}_1^2 = \mathcal{M}_2^2 = 2M^2$, Eq. (50) reduces to

$$G_1|_{\theta=\pi/2} = \frac{\exp[(M_1^2 + m_P^2)/(2M^2)](m_Q + m_s)}{M_1 f_A m_P^2 f_P} \hat{T}_A^{\text{OPE}}. \quad (76)$$

For $Q = c$, $P = D_s$, and $M_1 = m_{D_{s1}}$, the right-hand side is exactly the prefactor and axial OPE appearing in Eq. (4.4) of Colangelo, De Fazio, and Ozpineci. Therefore the relation is analytic, not merely a numerical similarity. Choosing $\theta = 0$ instead places the pure axial current in the second row of Eq. (45); it swaps the low/high labels and is not our $D_{s1}(2460)$ convention.

The numerical pure- A comparison is shown in Table 2.

Calculation	G [GeV $^{-1}$]	Γ [keV]
Colangelo et al., single J_A	-0.37 to -0.29	19 to 29
This project, Stage 2 at $\theta = 90^\circ$	-0.3978 to -0.3124	20.51 to 33.26
Interval overlap	-0.3700 to -0.3124	20.51 to 29.00

Table 2: Pure-axial benchmark. The remaining displacement is caused by the input set and convention choices, not by the state projector.

No intermediate angle is fitted here. Formally, at fixed residues, $G_1(\theta) = a \sin \theta + b \cos \theta$ with $a = f_A^{(1)} G_A^{(1)} / f_1$ and $b = f_B^{(1)} G_B^{(1)} / f_1$. A physical angle extraction requires the complete tensor-current amplitude and the angle-dependent physical residue $f_1(\theta)$; both are still incomplete.

10.5 Decay constants from the rotated two-point correlator

Introduce the transverse two-point matrix

$$\Pi^{\text{basis}} = \begin{pmatrix} \Pi_{AA} & \Pi_{AB} \\ \Pi_{AB} & \Pi_{BB} \end{pmatrix}. \quad (77)$$

The physical matrix is $\Pi^{\text{phys}} = R \Pi^{\text{basis}} R^T$, so that

$$\Pi_{11} = \sin^2 \theta \Pi_{AA} + \cos^2 \theta \Pi_{BB} + 2 \sin \theta \cos \theta \Pi_{AB}, \quad (78)$$

$$\Pi_{22} = \cos^2 \theta \Pi_{AA} + \sin^2 \theta \Pi_{BB} - 2 \sin \theta \cos \theta \Pi_{AB}, \quad (79)$$

$$\Pi_{12} = \sin \theta \cos \theta (\Pi_{AA} - \Pi_{BB}) + (\cos^2 \theta - \sin^2 \theta) \Pi_{AB}. \quad (80)$$

This is already the useful paper-level explicit form. If each basis correlator is decomposed as

$$\hat{\Pi}_{XY} = \int_{s_{\text{th}}}^{s_0} ds e^{-s/\mathcal{M}^2} \rho_{XY}^{\text{pert}}(s) + \sum_d \hat{\Pi}_{XY}^{(d)}, \quad XY = AA, BB, AB, \quad (81)$$

then substitution gives, without an unreadably long expansion,

$$\begin{aligned} \hat{\Pi}_{11} = & \int_{s_{\text{th}}}^{s_0} ds e^{-s/\mathcal{M}^2} \left[\sin^2 \theta \rho_{AA}^{\text{pert}} + \cos^2 \theta \rho_{BB}^{\text{pert}} + 2 \sin \theta \cos \theta \rho_{AB}^{\text{pert}} \right] \\ & + \sum_d \left[\sin^2 \theta \hat{\Pi}_{AA}^{(d)} + \cos^2 \theta \hat{\Pi}_{BB}^{(d)} + 2 \sin \theta \cos \theta \hat{\Pi}_{AB}^{(d)} \right], \end{aligned} \quad (82)$$

$$\begin{aligned} \hat{\Pi}_{22} = & \int_{s_{\text{th}}}^{s_0} ds e^{-s/\mathcal{M}^2} \left[\cos^2 \theta \rho_{AA}^{\text{pert}} + \sin^2 \theta \rho_{BB}^{\text{pert}} - 2 \sin \theta \cos \theta \rho_{AB}^{\text{pert}} \right] \\ & + \sum_d \left[\cos^2 \theta \hat{\Pi}_{AA}^{(d)} + \sin^2 \theta \hat{\Pi}_{BB}^{(d)} - 2 \sin \theta \cos \theta \hat{\Pi}_{AB}^{(d)} \right]. \end{aligned} \quad (83)$$

Writing $\hat{\Pi}_{ij}$ for the continuum-subtracted Borel invariants, the direct decay-constant sum rules would be

$$f_1^2(\theta) = \frac{e^{M_1^2/\mathcal{M}^2}}{M_1^2} \left[\sin^2 \theta \hat{\Pi}_{AA} + \cos^2 \theta \hat{\Pi}_{BB} + 2 \sin \theta \cos \theta \hat{\Pi}_{AB} \right], \quad (84)$$

$$f_2^2(\theta) = \frac{e^{M_2^2/\mathcal{M}^2}}{M_2^2} \left[\cos^2 \theta \hat{\Pi}_{AA} + \sin^2 \theta \hat{\Pi}_{BB} - 2 \sin \theta \cos \theta \hat{\Pi}_{AB} \right]. \quad (85)$$

The condition $\hat{\Pi}_{12} \simeq 0$ can be used to diagnose or determine the preferred mixing angle. At $\theta = \pi/2$, Eq. (84) becomes the pure axial-current result

$$f_1^2 = \frac{e^{M_1^2/\mathcal{M}^2}}{M_1^2} \hat{\Pi}_{AA}, \quad (86)$$

which is the decay-constant normalization used by Colangelo et al.

The qualification “would be” is important: in this project only the AA OPE below is explicit. A dedicated local calculation of ρ_{BB} , ρ_{AB} and their condensate terms has not yet been completed. Consequently Eqs. (82) and (83) are exact rotation formulae, but they are not yet three fully evaluated two-point OPEs.

For completeness, their explicit axial two-point sum rule, generalized by $m_c \rightarrow m_Q$, is

$$f_A^2 = \frac{e^{M_A^2/\mathcal{M}^2}}{M_A^2} \left\{ \frac{1}{8\pi^2} \int_{(m_Q+m_s)^2}^{s_0} ds \lambda^{1/2}(s, m_Q^2, m_s^2) \right. \\ \times \left[2 - \frac{m_Q^2 + m_s^2 + 6m_Q m_s}{s} - \frac{(m_Q^2 - m_s^2)^2}{s^2} \right] e^{-s/\mathcal{M}^2} \\ \left. + e^{-m_Q^2/\mathcal{M}^2} \left[\langle \bar{s}s \rangle \left(m_Q - \frac{m_Q^2 m_s}{2\mathcal{M}^2} + \frac{m_Q^3 m_s^2}{2\mathcal{M}^4} \right) - \frac{\langle \bar{s}\sigma_{\mu\nu} g_s G^{\mu\nu} s \rangle}{4} \frac{m_Q^3}{\mathcal{M}^4} \right] \right\}, \quad (87)$$

where $\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2ac - 2bc$. Notice that the mixed condensate occurs here in the two-point decay-constant sum rule even though it has not been included in our radiative three-point sum rule. For the charm channel the benchmark value is $f_A \simeq 0.225$ GeV.

10.6 What was actually obtained for f_1 and f_2

The numerical scripts used the reduced overlap model obtained after stripping the common pole and Borel factors,

$$f_1^2 = \sin^2 \theta f_A^2 + \cos^2 \theta f_B^2 + 2 \sin \theta \cos \theta \rho_{AB} f_A f_B, \quad (88)$$

$$f_2^2 = \cos^2 \theta f_A^2 + \sin^2 \theta f_B^2 - 2 \sin \theta \cos \theta \rho_{AB} f_A f_B. \quad (89)$$

For the D_s sector the value $f_1 = 0.345(17)$ GeV was used as a working anchor, so that

$$\rho_{AB} = \frac{f_1^2 - \sin^2 \theta f_A^2 - \cos^2 \theta f_B^2}{2 \sin \theta \cos \theta f_A f_B}, \quad (90)$$

and Eq. (89) then predicted f_2 . For the B_s sector no f_1 anchor was available; instead the model imposed $\Pi_{12} = 0$, which gives

$$\rho_{AB}^{\text{diag}} = -\frac{\sin \theta \cos \theta (f_A^2 - f_B^2)}{(\cos^2 \theta - \sin^2 \theta) f_A f_B}. \quad (91)$$

The saved central-angle scan summaries are

Sector and input	f_1 [GeV]	f_2 [GeV]	Status
D_s , legacy χ_s	0.344 [0.329, 0.363]	0.380 [0.339, 0.423]	f_1 anchor; f_2 inferred
D_s , lattice $f_{\gamma,s}^\perp$	0.345 [0.330, 0.364]	0.379 [0.338, 0.423]	f_1 anchor; f_2 inferred
$B_s(5830)$ closure, legacy χ_s	0.436 [0.380, 0.486]	0.218 [0.138, 0.292]	$\Pi_{12} = 0$ model
$B_s(5830)$ closure, lattice $f_{\gamma,s}^\perp$	0.443 [0.384, 0.492]	0.203 [0.121, 0.291]	$\Pi_{12} = 0$ model

Table 3: Median and 16–84 percentile intervals at $\theta = 35.3^\circ$. These are the values found in the existing scans, but they are not independent predictions of a completed $AA/BB/AB$ two-point OPE calculation.

For the paper, the safe presentation is therefore to show Eqs. (78)–(80), quote the overlap-model values with this qualification, and reserve the phrase “QCD sum-rule prediction of f_1, f_2 ” for the future explicit BB and AB calculation.

Finally, once G_i is normalized with the corresponding f_i , the decay width is

$$\Gamma(H_i \rightarrow P\gamma) = \frac{\alpha_{\text{em}}}{3} |G_i|^2 k_{\gamma,i}^3, \quad k_{\gamma,i} = \frac{M_i^2 - m_P^2}{2M_i}. \quad (92)$$

11 Inverse-denominator reduction

The projected expressions contain k^2 , $k \cdot p$, and $k \cdot q$. For the heavy-emission topology these are replaced by

$$k^2 = d_{H2} + m_Q^2, \quad (93)$$

$$k \cdot q = \frac{d_{H1} - d_{H2}}{2}, \quad (94)$$

$$k \cdot p = \frac{d_{H2} + m_Q^2 + p^2 - m_s^2 - d_{H3}}{2}. \quad (95)$$

For strange emission,

$$k^2 = d_{S1} + m_Q^2, \quad (96)$$

$$k \cdot p = \frac{d_{S1} + m_Q^2 + p^2 - m_s^2 - d_{S2}}{2}, \quad (97)$$

$$k \cdot q = p \cdot q + \frac{d_{S2} - d_{S3}}{2}. \quad (98)$$

Setting all three inverse denominators of a topology to zero extracts the coefficient of the genuine three-denominator triangle. Terms containing one or more d_i cancel propagators and are retained separately in the full machine output.

The axial triangle cores are

$$N_H^{A,\text{core}} = 4im_Q + \frac{4im_Q^3 - 4im_Q^2 m_s}{p \cdot q}, \quad (99)$$

$$N_s^{A,\text{core}} = -4im_s + \frac{4im_Q m_s^2 - 4im_s^3}{p \cdot q}. \quad (100)$$

The tensor-current cores are

$$N_H^{B,\text{core}} = \frac{8im_Q^2 - 4im_Q m_s}{m_Q + m_s} + \frac{4im_Q^2 p^2}{(m_Q + m_s)(p \cdot q)}, \quad (101)$$

$$N_s^{B,\text{core}} = \frac{-4im_Q m_s + 8im_s^2}{m_Q + m_s} + \frac{4im_s^2 p^2}{(m_Q + m_s)(p \cdot q)}. \quad (102)$$

These expressions refer only to the numerator traces of Eqs. (38)–(39); the charge, color, propagator, QED-vertex, and global Wick factors must still be attached.

12 Feynman parameterization of the hard triangles

For three denominators,

$$\frac{1}{D_1 D_2 D_3} = 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xD_1 + yD_2 + (1-x-y)D_3]^3}. \quad (103)$$

Equivalently one may use $x + y + z = 1$.

12.1 Heavy emission

Assign x, y, z to d_{H1}, d_{H2}, d_{H3} . Direct completion of the square gives

$$xd_{H1} + yd_{H2} + zd_{H3} = (k + xq - zp)^2 - \Delta_H, \quad (104)$$

where

$$\Delta_H = (x + y)m_Q^2 + zm_s^2 - xzp'^2 - yzp^2. \quad (105)$$

12.2 Strange emission

Assign x, y, z to d_{S1}, d_{S2}, d_{S3} . Then

$$xd_{S1} + yd_{S2} + zd_{S3} = [k - yp - z(p + q)]^2 - \Delta_s, \quad (106)$$

with

$$\Delta_s = xm_Q^2 + (y + z)m_s^2 - xyp^2 - xzp'^2. \quad (107)$$

The independent pairwise-invariant formula

$$\Delta = \sum_i x_i m_i^2 - \sum_{i < j} x_i x_j (r_i - r_j)^2 \quad (108)$$

reproduces Eqs. (105) and (107). The notebook checks both the completed-square and pairwise forms:

```
deltaH = (x + y) mQ^2 + z ms^2 - x z pPrime2 - y z p2;
shiftH = x q - z p;
heavyResidual = simplexReduce[
  x dH1 + y dH2 + z dH3
  - (SP[k + shiftH, k + shiftH] - deltaH)
];

deltaS = x mQ^2 + (y + z) ms^2 - x y p2 - x z pPrime2;
shiftS = -y p - z (p + q);
strangeResidual = simplexReduce[
  x dS1 + y dS2 + z dS3
  - (SP[k + shiftS, k + shiftS] - deltaS)
];
```

Both residuals vanish exactly.

13 Soft two-particle Fierz traces

After contracting the heavy quark and Fierz rearranging the strange bilocal, the Dirac factor is

$$T_i^X = \text{Tr}_D[\Gamma_P(\not{k} + m_Q)\Gamma_\mu^X\Gamma_i]. \quad (109)$$

The notebook evaluates the standard scalar, pseudoscalar, vector, axial, and tensor basis. The nonzero results for the axial current are

$$T_1^A = 4ik_\mu, \quad (110)$$

$$T_{\gamma_\rho}^A = -4im_Q g_{\mu\rho}, \quad (111)$$

$$T_{\sigma_{\rho\lambda}}^A = 4g_{\lambda\mu}k_\rho - 4g_{\mu\rho}k_\lambda. \quad (112)$$

For the tensor current,

$$T_{\gamma_\rho}^B = \frac{4i}{m_Q + m_s} [k_\mu p'_\rho - g_{\mu\rho}(k \cdot p')], \quad (113)$$

$$T_{\gamma_\rho \gamma_5}^B = -\frac{4}{m_Q + m_s} \epsilon_{\mu\rho k p'}, \quad (114)$$

$$T_{\sigma_{\rho\lambda}}^B = \frac{4m_Q}{m_Q + m_s} [g_{\lambda\mu}p'_\rho - g_{\mu\rho}p'_\lambda]. \quad (115)$$

All omitted entries vanish. Cyclically rotating the matrix chain gives the same tensor- and vector-basis traces for both currents.

14 Implemented OPE terms: exact status

It is not correct to describe every term below as independently completed in the new Mathematica restart. The existing project contains a mixture of published axial-benchmark terms, new exact symbolic traces, and tensor-current estimates. The present status is:

OPE contribution	Axial current A	Tensor current B
Perturbative photon from Q and s	Published spectral density implemented numerically; corrected traces independently reproduced	Corrected traces and triangle cores derived; diagonal double-dispersion spectral mapping implemented in Python
Local $e_Q \langle \bar{s}s \rangle$ term	Excluded in the Rohrwild result; retained only in the Colangelo comparison with m_s, m_s^2 corrections	Excluded in the Rohrwild result; no Colangelo J_B term invented
Electromagnetic three-particle DAs, $\mathcal{S}_\gamma, \mathcal{T}_4^\gamma$	Included through Eq. (65)	Included through Eq. (69) with the stated physical-residue numerator prescription
Twist-2 two-particle DA, $\chi \langle \bar{s}s \rangle \phi_\gamma$	Included in the numerical axial benchmark	Tensor trace and numerical trace-ratio contribution included
Twist-4 two-particle DAs, $\mathbb{A}$ and h_γ	Included in the axial benchmark	Included with the tensor trace ratio at the Borel saddle
Twist-3 two-particle DA, $f_{3\gamma} \psi^v$	Included in the axial benchmark	Included with the physical-residue trace ratio (it vanishes at the symmetric point for the default antisymmetric ψ^v integral)
Three-particle photon DAs, $\mathcal{S}, \tilde{\mathcal{S}}, \mathcal{T}_{1\dots 4}$	Full published $\mathcal{F}_1$ integral implemented numerically	Local- σ and $x_\alpha \gamma_\beta$ kernels included with the documented double-pole residue-matching prescription
Mixed condensate $\langle \bar{s}g_s \sigma \cdot Gs \rangle$	Not included in the radiative three-point sum rule	Not included
Vacuum gluon condensate $\langle G^2 \rangle$	Not included in the radiative three-point sum rule	Not included

Table 4: Term-by-term distinction between implemented results and incomplete parts of the independent derivation.

15 Verification summary and present boundary

Check	Result	Exact tests
Currents, Wick signs, traces, E1 cores, soft trace cyclicity	PASS	19/19
Standard vertex routing	PASS	4/4 momenta
Heavy- and strange-line Ward identities	PASS	2/2
Shifted and pairwise Feynman denominators	PASS	4/4
State rotation, OPE assembly, Colangelo limit, decay constant, and width	PASS	14/14
Sequential evaluation of the interactive notebook	PASS	38/38 cells

Table 5: Status of the independent Mathematica/FeynCalc restart.

The B_s channel uses the same symbolic expressions with $m_Q \rightarrow m_b$ and the appropriate heavy-quark charge. No second Dirac-trace derivation is needed.

The calculation deliberately stops here. The next independent tasks are:

1. independently audit the separately implemented $e_Q \langle \bar{s}s \rangle$ contribution and decide whether it is retained in the final numerical convention;
2. reproduce the axial two- and three-particle photon-DA terms from the explicit matrix elements rather than importing the final published sum rule;
3. perform the shifted loop integrals based on Eqs. (105)–(107);
4. derive the double discontinuity or an equivalent double-dispersion representation;
5. apply continuum subtraction and the double Borel transform;
6. reproduce the published axial-current spectral density as a normalization and sign benchmark;
7. only then derive and numerically implement the tensor-current hard spectral density.

The earlier Python spectral-density candidate is not used as an input to this restart and is not changed by the work documented here.

A Reproduction commands

From `papers/Ds_Bs_gamma/`, run

```
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/mathematica_current_wick_trace_audit.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/mathematica_correlator_routing_audit.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/mathematica_hard_loop_parameterization.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/mathematica_state_mixing_projection.wl
```

To regenerate the notebook from its source cells, run

```
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/build_dsbs_symbolic_notebook.wl
```